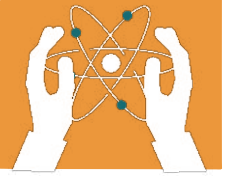


UNIT 7

REFLECTION, SHADOWS AND TRANSPARENT OBJECTS

Learning objectives:

- Understand light's interactions with surfaces, including reflection and shadow formation.
- Identify transparent, translucent, and opaque objects based on light passage.
- Recognize how shadow size changes with the object's distance from the light source.



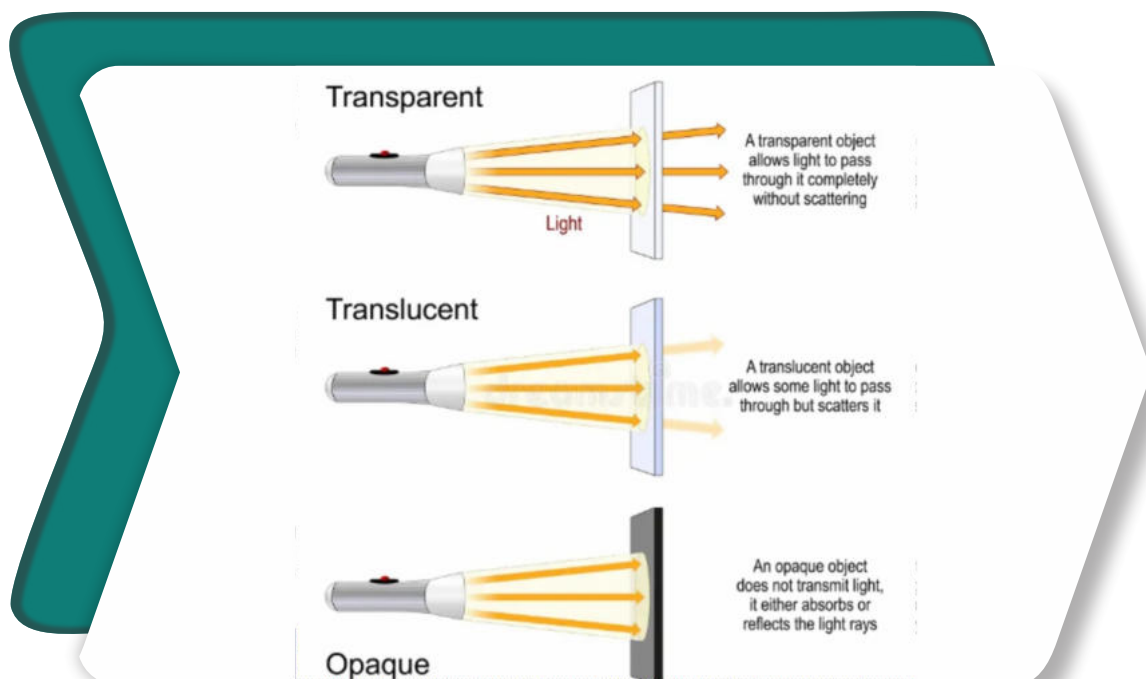
Light plays an essential role in how we see the world around us. It interacts with different surfaces and objects in fascinating ways. In this chapter, we will explore how light behaves when it reflects, how shadows are formed, and which objects allow light to pass.

How Light Interacts with Surfaces

Light behaves differently when it hits various surfaces or objects. Let's explore three main types of interactions:

Transparent, Translucent, and Opaque Objects

- **Transparent objects** allow light to pass through them completely, like glass.
- **Translucent objects** allow some light to pass through, but not all, like frosted glass.
- **Opaque objects** do not allow any light to pass through them, like wood or metal.



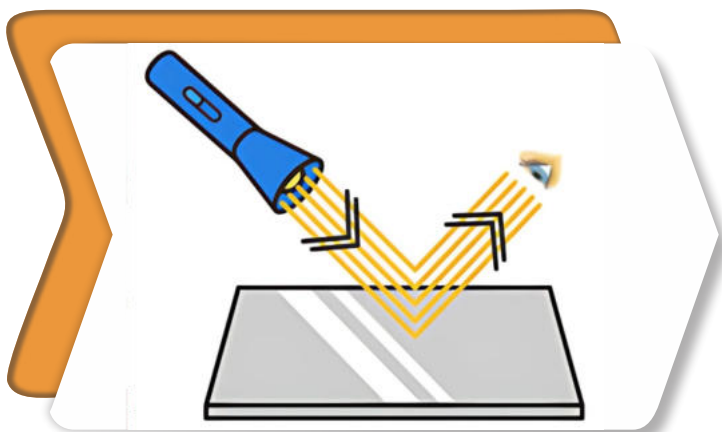


REFLECTION, SHADOWS AND TRANSPARENT OBJECTS

What happens when you hit the ball on wall?

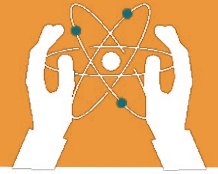
Reflection

Reflection happens when light bounces off a surface. Smooth and shiny surfaces, like mirrors, reflect light clearly. This is why we can see our face in a mirror.



Shadows

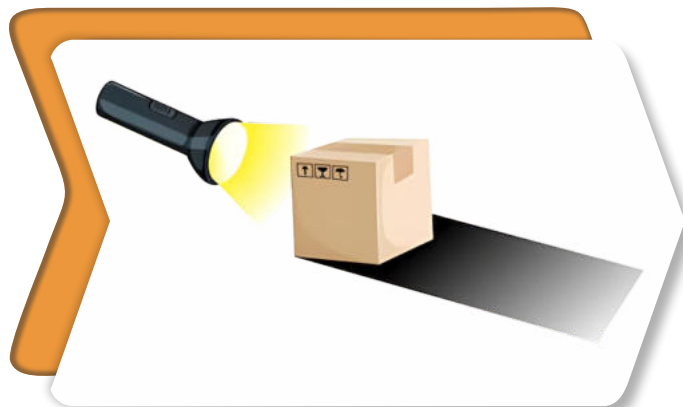
Shadows are formed when an opaque object blocks the path of light. Shadows always appear on the side opposite the light source. The size and shape of a shadow depend on the position of the object and the light.

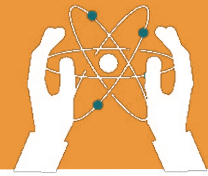


How Shadows Change Size?

The size of a shadow depends on how close the object is to the light source.

- If the object is close to the light, the shadow will be large.
- If the object is far from the light, the shadow will be smaller.



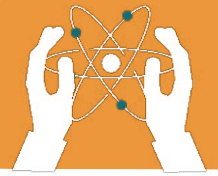


REFLECTION, SHADOWS AND TRANSPARENT OBJECTS

Fill in the blanks.

1. Light _____ off smooth surfaces like mirrors.
2. Shadows are formed when an object _____ the path of light.
3. Transparent objects allow _____ of the light to pass through them.
4. _____ objects do not let any light pass through.
5. The _____ of a shadow depends on the distance between the object and the light source.
6. Reflection happens when light _____ back from a surface.
7. Frosted glass is an example of a _____ object.
8. A shadow always forms on the _____ side of the light source.

REFLECTION, SHADOWS AND TRANSPARENT OBJECTS



Write “T” for True or “F” for False.

1. Opaque objects let all the light pass through.

2. Transparent objects block light completely.

3. Shadows are formed on the same side as the light source.

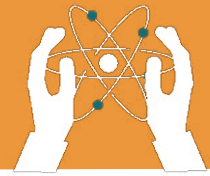
4. A mirror reflects light clearly.

5. A shadow gets bigger when the object is closer to the light.

6. Reflection occurs when light bounces off a surface.

7. Translucent objects allow some light to pass through.

8. Shadows appear darker on sunny days.



Choose the correct answer.

1. What happens when light hits a mirror?

- It passes through the mirror
 - It bounces back
 - It gets absorbed
 - It disappears
-

2. What kind of object blocks all light?

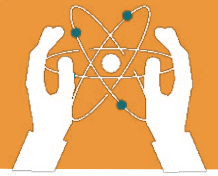
- Transparent
 - Translucent
 - Opaque
 - Reflective
-

3. Which object allows all light to pass through?

- Glass
 - Wood
 - Frosted glass
 - Metal
-

4. What determines the size of a shadow?

- Color of the object
- Distance from the light source
- Shape of the object
- Brightness of the light



5. What kind of surface reflects light clearly?

- Smooth and shiny
 - Rough and dull
 - Transparent
 - Translucent
-

6. Which of these is an example of a translucent object?

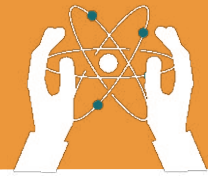
- Clear glass
 - Wood
 - Frosted glass
 - Steel
-

7. What do you need for a shadow to form?

- A transparent object
 - A light source and an opaque object
 - A mirror
 - A translucent object
-

8. If an object is far from the light, its shadow will be:

- Larger
 - Smaller
 - Brighter
 - Invisible
-



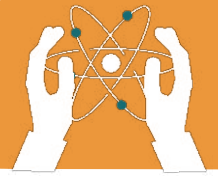
Answer the following questions.

- Q1. Write down the uses of light.**
- Q2. How does a shadow change in size based on the distance from the light source?**
- Q3. What happens when light interacts with transparent, translucent, and opaque objects?**
- Q4. What do you know about reflection?**
- Q5. Explain how shadows are formed.**
- Q6. Why do opaque objects create clear shadows, but translucent ones do not?**
- Q7. What is the main difference between frosted glass and clear glass in terms of light passage?**



Critical Thinking

Why do we see our reflection clearly in a mirror but not on a wall, even though both surfaces reflect light?



Create Shadows of Different Objects



ACTIVITY

- **Materials Needed:** A flashlight, transparent glass, frosted plastic sheet, and wooden block.
- **Steps:**
 1. Shine the flashlight on the glass to see light passing through.
 2. Shine it on the frosted plastic to notice partial light passing.
 3. Shine it on the wooden block to observe the shadow.
 4. Move the wooden block closer or farther from the flashlight to observe changes in shadow size.

Fun Fact

- It takes about 8 minutes 20 second for light to pass from sun to earth.
- Light always travels in a straight line.
- White light is made up of all colours of rainbow.